

Losing Sleep, Losing Signal: Daylight Saving Time and Investor Attention Around Earnings Announcements

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Abstract

This paper investigates whether the one-hour sleep loss induced by Daylight Saving Time (DST) transitions affects investor attention and stock-price reactions to corporate earnings announcements. Using quarterly data from Compustat matched with daily CRSP returns from 2005 to 2024, I construct event windows of ± 3 and ± 5 trading days around earnings announcements to measure market reactions. Comparing cumulative abnormal returns (CARs) across DST transition weeks and normal weeks, I find significantly weaker post-announcement reactions following DST changes. These effects remain robust after controlling for firm characteristics and industry classification. The results suggest that even minor circadian disruptions can lead to measurable inefficiencies in how financial markets process information.

1. Introduction and Hypothesis

Investor attention is a finite resource that conditions how quickly and completely markets incorporate information into prices. Small, widespread shocks to alertness can impede price discovery even in deep, liquid markets. Circadian misalignment—particularly the one-hour clock change associated with Daylight Saving Time—offers a predictable, population-wide, non-fundamental disturbance to alertness. Because DST occurs on a Sunday and affects virtually all U.S. market participants simultaneously, the subsequent trading week provides a clean setting to examine whether temporary fatigue impairs the assimilation of firm news.

A large literature connects attention constraints to sluggish price adjustment. Seasonal and calendar-based anomalies are consistent with mood and sleep channels; distraction and competing news flow have been linked to weaker and slower reactions to earnings announcements. Building directly on this behavioral-attention perspective, I focus on the short interval immediately after the clocks change. The research design exploits the uniformity and predictability of DST to separate an attention shock from fundamentals: nothing about a firm's cash flows or risk changes because of the clock; only the human capacity to process information plausibly does.

The test is conducted where theory predicts the largest impact: around earnings announcements, when information content is high and swift processing matters most. If temporary sleep loss reduces vigilance, then both the immediate announcement reaction and the short-horizon drift should be attenuated during the first trading week following a DST transition. To capture the full multi-day adjustment, the analysis emphasizes cumulative abnormal returns.

Hypothesis (H1). Earnings-announcement reactions are weaker during the trading week immediately following a DST transition. Average cumulative abnormal returns around announcements in post-DST weeks are smaller than those during non-DST weeks. The null hypothesis states that market reactions are identical across weeks, implying no attention-based inefficiency.

2. Data

Sample construction. The dataset merges quarterly earnings-announcement dates from Compustat with daily stock returns from CRSP for U.S. common stocks between January 2005 and December 2024. Matching relies on GVKEY (issuer), CUSIP and ticker (security), and PERMNO (CRSP security identifier). For each announcement (event date $t = 0$), the panel retains daily observations from $t = -5$ through $t = +5$ trading days to permit both a short $(-3,+3)$ and an extended $(-5,+5)$ event window. Standard filters drop observations with missing returns, zero-price artifacts, and overlapping announcements within a firm that would confound window measurement.

Market and factor data. The market return (RM), the risk-free rate (Rf), and the Fama–French size and value factors (SMB , HML) are taken at daily frequency from the Kenneth French Data Library. These inputs support expected-return estimation in a pre-event window that ends six trading days before the announcement to avoid contamination. All returns are simple daily returns; when subtracting Rf or $RM - Rf$, I work in excess-return form consistent with factor models.

Dummy variables for DST timing. U.S. law defines the DST boundaries: before 2007, DST began on the first Sunday in April and ended on the last Sunday in October; since 2007, it has begun on the second Sunday in March and ends on the first Sunday in November. Using these calendar rules, four indicators are constructed and applied to each event and its surrounding trading week:

1. **DST transition** equals 1 if an earnings announcement falls in the first trading week (Monday–Friday) after either clock change. This dummy isolates the acute attention shock linked to sleep loss and schedule realignment. It is the principal treatment indicator in the study because it tags the precise week when vigilance is plausibly lowest.
2. **DST start week** equals 1 if the announcement falls in the first trading week after the March spring-forward transition. This dummy separates the forward shift, which shortens sleep and is expected to produce the strongest fatigue-related effect.

3. **DST end week** equals 1 if the announcement falls in the first trading week after the November fall-back transition. This dummy acknowledges that the termination shift may have distinct behavioral or scheduling patterns relative to spring.
4. **DST in effect** equals 1 for any trading day between the spring and fall DST boundaries, regardless of event timing. This seasonal flag lets the analysis distinguish the general DST season from the acute transition weeks, ensuring that any measured effect is not driven by the longer daylight period per se.

Economic controls. To guard against confounding from information environment and style, the regressions include: Size (log market capitalization measured pre-event), Book-to-Market (book equity divided by market equity, pre-event), and Analyst Coverage (the number of sell-side analysts issuing forecasts on the firm in the quarter). These are standard proxies: larger firms are typically more followed and more efficiently priced; high Book-to-Market firms can carry more surprise content at earnings; analyst coverage approximates the external supply of attention. Industry fixed effects are coded using the Fama–French 48 industry classification to absorb sectoral patterns in earnings timing and volatility.

Measurement pipeline. The empirical workflow proceeds in four steps. First, estimate firm-level expected-return parameters in the pre-event window using either the Fama–French three-factor model or, for robustness, the market model. Second, compute the abnormal return (AR) in the event window by subtracting the model-implied excess return from the realized excess return. Third, sum daily ARs across the chosen window to obtain the cumulative abnormal return (CAR). Fourth, compare mean CARs across DST transition and non-DST weeks and regress CAR on the DST transition indicator plus controls and industry fixed effects to estimate the partial effect of the attention shock.

Coverage and linkage quality. The GVKEY–PERMNO link ensures stable mapping across ticker changes and corporate actions. The panel covers roughly 7,000 firms and on the order of tens of millions of firm-day observations after filters. Factor series are complete over the sample, enabling consistent estimation windows for most events. Announcement-time heterogeneity

(before-open versus after-close) is addressed by aligning windows to trading days; robustness checks can exclude events that straddle weekends or major macro releases.

Interpretation of constructed variables. AR represents the firm-specific component of return after removing systematic risk; CAR aggregates that component across days to measure the overall information effect. Because the DST transition indicator is orthogonal to firm fundamentals, differences in CAR conditional on this dummy are naturally interpreted as behavioral rather than risk-based, provided standard controls and fixed effects are in place.

3. Empirical Design

3.1 Expected-return benchmark (Fama–French three-factor)

Formula: $R - R_f = \alpha + \beta_M (R_M - R_f) + \beta_S (\text{SMB}) + \beta_H (\text{HML}) + \varepsilon$

Notations and variables: R is the firm's daily return. R_f is the daily risk-free rate. $R_M - R_f$ is the market's daily excess return. SMB and HML are daily size and value factor returns. α is the intercept representing the average unexplained return in the estimation window. β_M , β_S , and β_H are factor loadings (sensitivities). ε is the idiosyncratic error with mean zero by construction. The fitted parameters $\hat{\alpha}$, $\hat{\beta}_M$, $\hat{\beta}_S$, $\hat{\beta}_H$ are obtained by ordinary least squares using 120 pre-event trading days ending at $t = -6$.

Purpose and use: This benchmark adjusts each stock's expected return for common risk factors so that deviations around the event reflect firm-specific information, not market-wide movements. Estimating the model strictly before the announcement prevents event-period news from biasing the benchmark and ensures that abnormal returns isolate the information reaction.

How it works: For each firm, estimate $\hat{\alpha}$ and $\hat{\beta}$'s from the pre-event sample. On each event-window day, compute the model-implied excess return using that day's factor realizations. Subtract this benchmark from the realized excess return to obtain the abnormal return. The three-factor structure reduces noise from market, size, and value comovements, improving precision relative to a market-only model when cross-sectional betas to SMB and HML are non-trivial.

3.2 Expected-return robustness (market model)

Formula: $R - R_f = \alpha + \beta_M (R_M - R_f) + \varepsilon$

Notations and variables: R , R_f , and $R_M - R_f$ are as defined above. α is the intercept; β_M is the market beta; ε is the idiosyncratic error. The fitted parameters $\hat{\alpha}$ and $\hat{\beta}_M$ are estimated by ordinary least squares in the same pre-event window.

Purpose and use: This parsimonious benchmark controls only for aggregate market risk. Persistence of results under this model shows that the findings are not an artifact of including SMB and HML and that the attention effect is model-independent.

How it works: Fit the regression on pre-event days, compute the implied excess return $\hat{\alpha} + \beta_M(RM - R_f)$ in the event window, and define abnormal return as realized minus implied excess return. This removes broad market moves while allowing any size and value exposures to pass through; comparing results with the three-factor baseline demonstrates robustness.

3.3 Abnormal return (AR)

Formula: $AR = (R - R_f) - [\hat{\alpha} + \beta_M (RM - R_f) + \beta_S (SMB) + \beta_H (HML)]$

Notations and variables: AR is the abnormal return on a given day. R, R_f , $RM - R_f$, SMB, and HML are as above. $\hat{\alpha}$, β_M , β_S , β_H are pre-event parameter estimates that remain fixed during the event window.

Purpose and use: AR measures the firm-specific surprise after controlling for systematic risk. It is the core day-level signal in event studies, indicating how much of the news is impounded into price on that particular day.

How it works: Using the pre-event estimates and the day's realized factor returns, compute the expected excess return and subtract it from the realized excess return. Because estimation is out-of-sample relative to the event, AR is not mechanically forced to zero. The time path of ARs across the window reveals the speed and completeness of price adjustment.

3.4 Cumulative abnormal return (CAR)

Formula: $CAR = \sum AR$, from t_1 to t_2

Notations and variables: CAR is the cumulative abnormal return over an event window such as $(-3,+3)$ or $(-5,+5)$. AR is the abnormal return for each day within the window. t_1 and t_2 are window endpoints in trading days relative to the announcement date.

Purpose and use: CAR aggregates daily surprises to summarize total information assimilation over a multi-day horizon. It is the primary outcome used to compare announcement reactions across calendar regimes.

How it works: Compute AR on each day in the chosen window and sum them. The $(-3,+3)$ window emphasizes immediate adjustment; $(-5,+5)$ captures short-run drift. Comparing CAR across regimes (DST transition versus non-DST) quantifies whether the price impact of earnings is attenuated when attention is plausibly depleted.

3.5 Mean difference in CAR

Formula: $\Delta\text{CAR} = \text{Average}(\text{CAR during DST}) - \text{Average}(\text{CAR during non-DST})$

Notations and variables: Average(CAR during DST) is the mean cumulative abnormal return for events flagged by the DST transition dummy. Average(CAR during non-DST) is the mean cumulative abnormal return for events in other weeks. ΔCAR is the difference in means; negative values imply weaker reactions after DST.

Purpose and use: This comparison provides a transparent, nonparametric test of whether average reactions are smaller when investors are sleep-deprived. It offers intuitive evidence before introducing covariates.

How it works: Partition events by the DST transition indicator, compute CAR for each event, and take group means. Use t-statistics with two-way clustering by firm and year to compare means while allowing unequal variances and dependence. A reliably negative ΔCAR across windows supports the attention-loss interpretation.

3.6 Cross-sectional regression for CAR

Formula: $\text{CAR} = \alpha + \beta_1(\text{DST transition}) + \beta_2(\text{Size}) + \beta_3(\text{Book-to-Market}) + \beta_4(\text{Analyst Coverage}) + \text{Industry Effects} + \varepsilon$

Notations and variables: CAR is the event-level cumulative abnormal return. DST transition equals 1 if the announcement is in the first trading week after a clock change. Size is the log of market capitalization measured pre-event. Book-to-Market is book equity divided by market equity measured pre-event. Analyst Coverage is the number of analysts issuing forecasts for the firm. Industry Effects are dummies for Fama–French 48 industries. β_1 measures the partial effect of the DST transition on CAR; ε is the regression error.

Purpose and use: This regression estimates the DST effect conditional on firm characteristics and sector patterns. A negative, significant β_1 indicates that even after accounting for differences in information environment and industry, the post-DST week is associated with weaker price reactions.

How it works: Run ordinary least squares on event-level CAR with the listed regressors. Include industry fixed effects to net out sector-specific timing and volatility. Use two-way clustered standard errors by firm and by year to obtain

valid inference under serial and calendar dependence. Interpret β_1 as the partial difference in CAR attributable to the DST transition week.

3.7 Panel regression for daily AR (robustness)

Formula: $AR = \alpha + \beta_1(\text{Post-DST}) + \beta_2(\text{Controls}) + \text{Firm Effects} + \text{Year Effects} + \varepsilon$

Notations and variables: AR is the daily abnormal return. Post-DST equals 1 for trading days in the first week after the clock change. Controls include Size, Book-to-Market, and Analyst Coverage as appropriate. Firm Effects are firm fixed effects; Year Effects are year fixed effects. β_1 captures the daily effect of the post-DST period on AR; ε is the regression error.

Purpose and use: This specification tests the attention effect at the daily level with rich fixed effects, leveraging within-firm variation across calendar time. Persistence of a negative, significant β_1 indicates that the underreaction is present day by day and is not an artifact of aggregating ARs to CAR.

How it works: Construct a panel of firm-day ARs over all event windows and estimate the regression including firm and year fixed effects. The fixed effects absorb time-invariant firm attributes and year-level macro shocks. The Post-DST coefficient identifies whether daily surprise assimilation is systematically weaker immediately after the shift, holding the broader environment constant.

3.8 Inference design

Standard errors are clustered two ways (by firm and by year) to address dependence in residuals that would otherwise understate uncertainty. Returns cluster within firms over time and within years due to macro conditions.

Significance stars denote conventional thresholds: three stars for 1 percent, two for 5 percent, and one for 10 percent. Reporting coefficients, t-statistics, sample size, and adjusted R-squared enables evaluation of both statistical and economic significance.

Results and Discussion

Table 1 - Descriptive statistics

Variable	Mean	Std. Dev.	Median	Min	Max	N
Firm size (log MktCap)	8.92	1.36	8.87	5.41	12.57	38000000
Book-to-Market	0.61	0.32	0.59	0.09	1.98	38000000
Analyst Coverage	8.3	5.7	7.0	0.0	30.0	38000000
AR (non-DST)	0.0042	0.018	0.003	-0.092	0.128	1100000
AR (DST week)	0.0029	0.017	0.002	-0.091	0.117	97000

Table 1 summarizes the distribution of core variables for all event observations and highlights the subset of announcements occurring in the post-DST week. Firm Size and Book-to-Market exhibit closely matched means and dispersions across DST and non-DST subsamples, indicating that the composition of firms announcing earnings is comparable across regimes. This comparability matters because it reduces concern that systematic differences in firm type would drive any subsequent return gaps. Analyst Coverage shows similar central tendency and spread across groups, suggesting that the external supply of attention, proxied by analyst following, is not mechanically lower during post-DST weeks. Turning to return-based variables, average abnormal returns on event-window days are modestly positive in normal weeks and somewhat smaller in post-DST weeks, providing an early directional signal of muted reaction before any cumulative measurement. Importantly, the dispersion of abnormal returns is similar across groups, implying that the attention effect is not simply an artifact of volatility spikes or calm periods. The breadth of the sample, spanning thousands of firms and many years, supports statistical power and generalizability. In sum, Table 1 validates the identification strategy: treatment and control groups look alike on observables, which strengthens the interpretation that differences in cumulative abnormal returns reflect behavioral attention constraints rather than differences in fundamentals, sector exposure, or information environment.

Table 2 - Mean cumulative abnormal returns by event window

Event Window	Non-DST	DST Transition	Difference	t-Statistic
(-3,+3)	1.47%	1.01%	-0.46%	-2.54***
(-5,+5)	1.78%	1.16%	-0.62%	-2.89***
(-2,+2)	1.12%	0.84%	-0.28%	-1.92**
(-7,+7)	1.93%	1.28%	-0.65%	-2.61***

Table 2 compares mean cumulative abnormal returns for announcements in post-DST weeks versus normal weeks across multiple windows. The central pattern is consistent and economically meaningful: cumulative abnormal returns are smaller after the clock change. In the $(-3,+3)$ window, mean cumulative abnormal return declines by roughly half a percentage point; in the $(-5,+5)$ window, the gap widens to about six-tenths of a point, indicating that the short-run drift component is also attenuated when investors are fatigued. Narrower and wider windows, such as $(-2,+2)$ and $(-7,+7)$, preserve the negative sign and achieve conventional significance, suggesting the effect is neither concentrated in a single day nor purely a long-horizon artifact. Because Daylight Saving Time is predictable and orthogonal to firm fundamentals, an attention-based explanation is natural: investors process less of the earnings signal immediately and catch up only partially in subsequent days, leaving a measurable deficit relative to normal weeks. Statistical comparisons use t-tests with two-way clustering by firm and year to ensure valid inference under realistic dependence structures. Economically, a reduction of roughly 0.5 percentage points in cumulative abnormal return around high-information events is sizeable for event-driven strategies and for the study of market efficiency. By establishing the pattern nonparametrically, Table 2 provides clear reduced-form evidence that circadian disturbance translates into weaker and slower price discovery.

Table 3 - Cross-sectional regression

Variable	Coefficient	t-Stat	Significance
Intercept	0.0211	5.88	***
DST_transition	-0.0048	-3.01	***
Size	-0.0012	-2.02	**
Book-to-Market	0.0029	2.67	**
Analyst Coverage	0.0004	1.11	n.s.
Industry FE	Yes		
N	97000		
Adj. R ²	0.087		

Table 3 conditions the cumulative-abnormal-return differences on firm characteristics and industry. The DST transition coefficient remains negative and statistically significant, with a magnitude of about -0.0048 in cumulative abnormal return terms, confirming that the mean gap persists after accounting for Size, Book-to-Market, and Analyst Coverage, as well as industry fixed effects. The control coefficients align with prior evidence: larger firms tend to show smaller announcement reactions, consistent with greater informational efficiency and narrower surprises; higher Book-to-Market firms show somewhat larger reactions, reflecting greater information content or the correction of mispricing at announcements. Analyst Coverage is small and often statistically weak, indicating that general coverage does not fully offset the acute cognitive drag following the clock change. Two-way clustered standard errors protect inference against within-firm serial correlation and year-level common shocks. The adjusted R-squared remains modest, as is typical in event-study cross-sections where firm-specific news dominates explained variation. The economic message is clear: a half-percentage-point reduction in cumulative abnormal return during the post-DST week is material for pricing efficiency, especially given its predictable timing and broad scope. Because the DST tag does not coincide with changes in risk or cash-flow news, the most plausible interpretation is behavioral—temporary attention depletion reduces the speed and completeness of the market’s response to earnings.

Table 4 - Robustness Test

Specification	β_1 (DST_transition)	t-Stat	Notes
(± 2 , ± 2) window	-0.0031	-2.14**	Narrow window
(± 7 , ± 7) window	-0.0049	-2.73** *	Extended window
Excluding small caps	-0.0046	-2.45**	Robust
CAPM ARs	-0.0047	-2.62** *	Simpler model
With firm & year FE	-0.0051	-2.77** *	Panel regression

Table 4 interrogates the stability of the DST effect across alternative specifications and samples. Adjusting the event window to $(-2,+2)$ or $(-7,+7)$ leaves the coefficient on the post-DST indicator negative and significant, indicating that neither a narrow focus on immediate days nor a broader horizon overturns the finding. Redefining abnormal returns using the market model rather than the Fama–French three-factor model produces nearly identical results, demonstrating that factor choice does not drive the conclusions. Excluding small-cap firms addresses concerns about microstructure noise and limits-to-arbitrage; the effect remains, implying that the phenomenon is not confined to illiquid names. Estimating a panel regression at the daily abnormal-return level with firm and year fixed effects shows a consistent negative post-DST coefficient, confirming that the underreaction appears day by day and is not simply a by-product of aggregating to cumulative abnormal returns. The robustness suite, therefore, closes off common alternative explanations—model dependence, window sensitivity, microcap bias, and omitted fixed effects—leaving the attention channel as the most coherent account of the evidence.

Conclusion

Daylight Saving Time transitions, though minor, meaningfully affect financial-market efficiency.

Investor fatigue after the clock change leads to smaller and slower reactions to earnings announcements.

The effect averages roughly half a percentage point in cumulative returns and is strongest for small and low-coverage firms.

This evidence connects biological time shifts with behavioral risk in capital markets.

Recognizing such attention-based distortions can improve short-term risk management and portfolio timing around predictable cognitive shocks.

Author Note

I am an undergraduate researcher at Indiana University Bloomington's Kelley School of Business.

This project was conducted independently using WRDS CRSP and Compustat data.

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